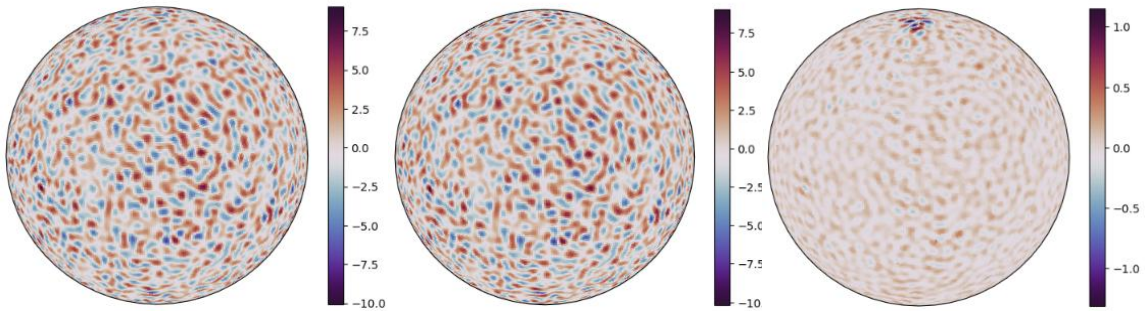
Loss Function:
Relative L2
Loss Function

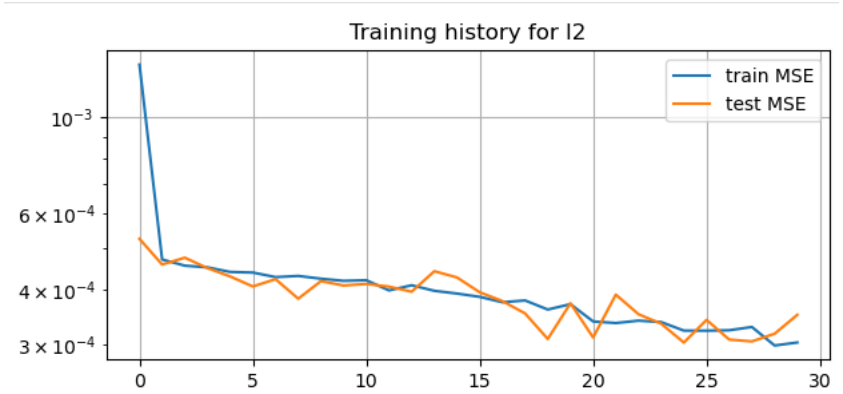
$$\sqrt{\frac{\sum_{i=1}^{n}\left(y_{target,i} - y_{predicted,i}\right)^2}{\sum_{i=1}^{n} y_{target,i}^2}}$$

- Sum of all errors normalized by sum of all targets.
- Relative: Allow scale invariance (dimensionless nature) and interpretable signal-to-error ratio.

Spatial l2 loss (weighs by quadrature weights)

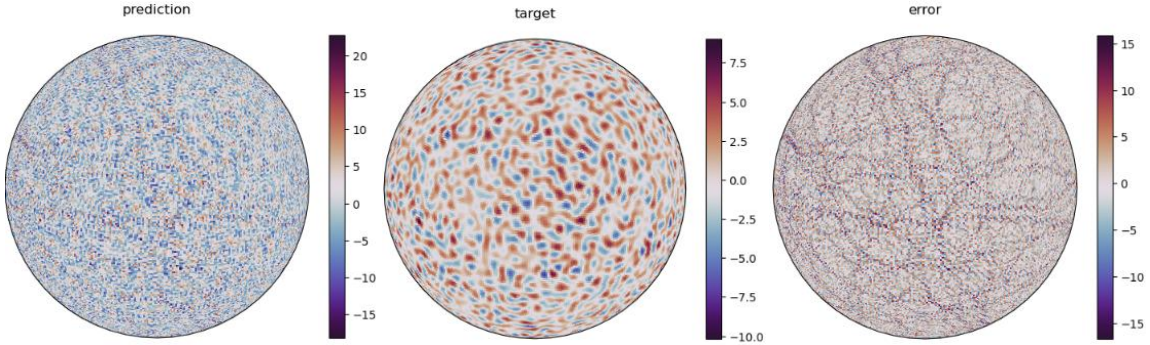
Prediction Target Error

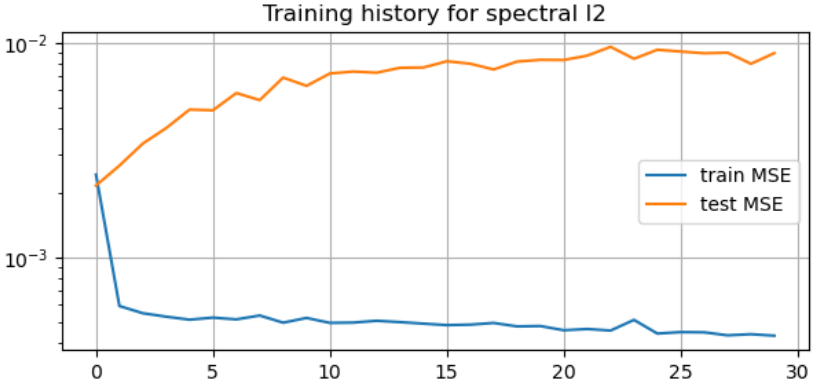




No gap between test/train loss curves and reasonable magnitude for loss. Test hop around train curve due to optimizer behavior and finite-sample noise.

Spectral L2 loss (same weighing on all spectral frequencies)





Overfit due to equal weights to physical low frequencies and erratic noisy high frequency modes.

Next Steps/Challenge: Retrain with weighted spectral L2 and a larger parameter set; explore autoregression, potentially including a forcing as an additional input.